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AirSense: ML-Driven Platform for Real-Time Air Quality Prediction Using a Low-Cost IoT Device

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***Abstract*—Air pollution is a leading health threat in rapidly growing urban areas, particularly Dhaka, Bangladesh, where stationary monitors are costly and sparsely deployed. We present AirSense, an end-to-end platform that combines low-cost mobile IoT sensing with Machine Learning (ML) analytics to generate high-resolution, real-time air-quality information. The system uses public-transport buses as mobile sensing nodes, equipping them with $PM_{2.5}$ sensors, GPS, and microcontrollers to collect geotagged spatio-temporal measurements across urban, semi-urban, and peri-urban areas. Collected data are ingested into a backend that employs Random Forest (RF) and Feedforward Neural Network (FNN) models for pollutant forecasting, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) for hotspot detection via spatial clustering. The platform uses sensor fusion and spatio-temporal interpolation to turn sparse, dynamic measurements into continuous pollution maps along important bus routes. It also gives solid short-term forecasts, which typical static monitors can't do. Interactive web dashboards surface real-time insights and system diagnostics for public-health authorities and policymakers. A field deployment on Dhaka buses demonstrates the platform's practicality and effectiveness. Future work will explore advanced predictive models, additional sensors, and enhanced communication protocols to improve autonomy and scalability.**

***Index Terms*—IoT, Air Pollution Monitoring System, Mobile Air Quality Monitoring, PM2.5 Sensors, Predictive Analytics**

I. Introduction

Air pollution is a critical global issue, particularly in densely populated urban areas such as Dhaka, Bangladesh. It is responsible for approximately 3 million deaths annually from outdoor exposure and 4.3 million deaths from household air pollution, including smoke from cooking with solid fuels [1]. To combat this, smart cities now rely on real-time data and monitoring, with the Internet of Things (IoT) playing a crucial role in enabling continuous data collection. Furthermore, Low-cost IoT systems are critical in developing countries, particularly as industrial parks increase pollution problems. In recent years, rapid industrial growth has caused significant environmental harm, prompting the World Health Organization (WHO) to set air quality guidelines for pollutants like ozone, NO_2, and SO_2 [2]. Traditional air quality monitoring systems are often costly and difficult to scale; in contrast, IoT-based solutions provide a more affordable, scalable, and real-time alternative [3]. Urban air pollution primarily originates from vehicles and industrial activities, with harmful pollutants, e.g. Particulate Matter $PM_{2.5}$ ($\leq 2.5\,\mu m$) and PM_{10} ($\leq 10\,\mu m$), smog, CO (Carbon Monoxide), and NO_2 (Nitrogen Dioxide) posing serious health risks. Additional contributors include fires, construction, agriculture, mining, and factory waste [4].

Our project's goal is to reduce health harm from traffic-generated air pollution by proposing an affordable, portable IoT system for near-real-time monitoring. To safeguard against data loss, the system securely stores all information on a local SD card [4]. Filtered data is sent to a backend analytics pipeline where advanced machine learning algorithms like Random Forests (RF), Density-Based Spatial Clustering of Applications with Noise (DBSCAN), clean, feature-engineer, and process it. The UI delivers predictions, spatial heatmaps, and system diagnostics to monitor model and sensor health. The architecture also provides automatic triggers for diagnostics and maintenance to improve reliability. This cost-effective (50–60$) device combines Arduino technology with cloud connectivity, enabling scalability on public transit routes and offering a breakthrough in affordable, continuous urban air quality monitoring.

The key contributions of this research are:

- The design and practical validation of AirSense, a low-cost (approx. $53) and scalable mobile sensing platform deployed on public buses.
- The development of a high-accuracy (96.8%) and computationally-efficient FNN for real-time pollutant forecasting, which outperformed LSTM.
- The successful application of DBSCAN clustering to mobile data to identify and map real-world pollution hotspots along major traffic corridors in Dhaka.

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II. Related Works

Recent research looks into machine learning, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM), for predicting environmental and climate changes. This RNN architecture and LSTM algorithm are well-suited to processing sequential data, making them useful for real-time environmental forecasting [5]. Their accuracy improves as the dataset size increases. IoT and Wi-Fi technologies provide dependable data collection and real-time transmission, even in remote places [6]. Data pre-processing, such as eliminating noise and outliers, enhances analytical accuracy and consistency [7]. M5P decision trees are also useful for real-time environmental modeling, processing huge amounts of information effectively under dynamic situations such as changing air pollution levels [8]. The popularity of IoT-based air pollution monitoring systems is rising because they can measure pollutant emissions from various sources within industrial processes to ensure compliance with regulations. As a result, these systems clarify the sources of pollution and allow industries to monitor pollution in real time and take corrective actions to minimize emissions and have a lower environmental impact. More versions of IoT technology are being used for air quality monitoring purposes in urban spaces, industrial areas, and highways, while monitoring continuously in an area where fixed monitoring stations could not be placed, or might be costly to do so [9]. IoT-based systems also help authorities monitor pollution in densely populated urban areas and determine pollution hotspots, providing them with the necessary data to make pollution interventions more effective and develop policies to address pollution [10]. The data collected from IoT systems can also help local governments develop a strategy and implement a plan to reduce pollution and protect public health [11]. Unlike current systems, we use local busses as mobile air-quality monitoring stations and public transport vehicles to collect real-time pollution data in urban, semi-urban, and rural areas. Its mobility overcomes the cost and limited coverage of permanent monitoring stations, providing significant coverage. Our buses are equipped with $PM_{2.5}$ and GPS sensors as well as microcontrollers that provide continuous monitoring and store data locally if connectivity is lost. The IoT-backed software continuously monitors industrial sites, high-traffic roadways, and sensitive sites, such as hospitals and schools, for pollution hotspots.

A. High-Level Design: A Multi-Layered Architecture

III. System Design and Methodology

AirSense adopts a constructive research methodology aimed at the design, development, and validation of a scalable and portable full-stack air quality monitoring and prediction platform for urban environments. The system follows a structured pipeline that integrates IoT-based mobile sensing, cloud-enabled data acquisition, and AI-powered analytics. Engineered for robust performance in dynamic urban settings, the platform enables live data streaming, spatial-temporal analysis, and automated alerts to support informed decision-making. An architectural overview of the system is presented in Fig. 1, highlighting key components such as mobile sensor nodes, cloud data pipelines, analytics modules, and an interactive dashboard.

A. System Architecture and Technology Stack

The proposed platform is an end-to-end air quality monitoring and prediction system that integrates mobile sensing, intelligent analytics, and web-based visualization. As shown in Fig. 1, the system is divided into three primary layers: *IoT Data Acquisition*, *Backend Services and Analytics*, and *UI Presentation*. Each layer is designed to work in unison to provide scalable, real-time environmental monitoring across dynamic urban environments.

1) IoT Data Acquisition Layer: This layer is composed of mobile IoT nodes installed on public transport vehicles. Each node integrates a NodeMCU MEGA WiFi R3 microcontroller, a $PM_{2.5}$ dust sensor, a suite of MQ-series gas sensors (e.g., MP135, MP503), a GPS module (UBlox NEO-6M), and an SD card module. Data is collected from the environment in real time and filtered locally before being stored or transmitted. The GPS module allows for accurate geo-tagging, while the SD card ensures persistent local logging when internet connectivity is unavailable. The hardware prototype is shown in Fig. 2.

2) Backend Services and Analytics Layer: Filtered data from the IoT nodes is ingested into the backend system, where it undergoes cleaning and transformation. Feature engineering extracts temporal and spatial features to enhance model performance. The analytical engine comprises three core models: an RF classifier for pollutant prediction, an FNN for real-time multi-output forecasting, and DBSCAN for spatial clustering of pollution hotspots. Processed data and model outputs are stored in structured formats and served via RESTful API endpoints.

3) UI Presentation Layer: The web-based dashboard, developed using Next.js and TypeScript, enables users to interact with live and historical pollution data. It supports multiple data views, including charts, heatmaps, and system diagnostics. Users can explore predictive, spatial, and explanatory insights through dynamic visual components. This layer empowers city planners, researchers, and citizens to visualize and respond to air quality issues effectively, as shown in Fig. 3.

B. Component-Level Breakdown

Each mobile monitoring device is a carefully integrated assembly of hardware components designed for portability, robustness, and low power consumption. To minimize interference from the vehicle's own exhaust, the sensor units were mounted on the front roof of each bus. Table I presents the detailed cost breakdown of the system components.

1) Processing Core: An Arduino MEGA WiFi R3 (with ESP8266) manages all sensor interactions, data processing, and communication. Its low power consumption makes it suitable for mobile environments.

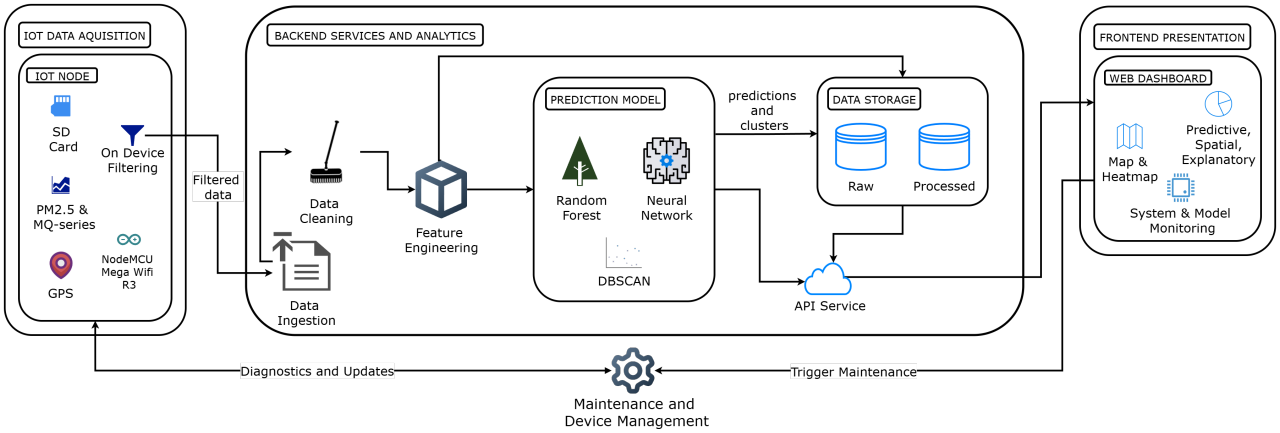


Fig. 1: System architecture depicting IoT data acquisition, AI-powered analytics, and interactive web-based visualization.

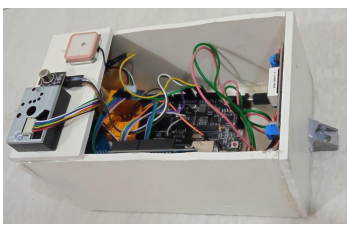


Fig. 2: Prototype after Hardware integration

TABLE I: Hardware Component Cost Analysis

Component	Price (BDT)	Price (USD)
Arduino MEGA 2560 with WiFi	1890	$15.12
PM2.5 Optical Smoke Detector	1350	$10.80
UBlox NEO-6M GPS Module	599	$4.79
Mini Data Logger Module	550	$4.40
Arduino MicroSD Card Module	75	$0.60
MP 135 / MP 503 Air Quality Sensor	590	$4.72
LiPo Battery 1200 mAh 11.1V	1090	$8.72
B3 Pro LiPo Balance Charger	499	$3.99
Total	**6643**	**$53.14**

2) Pollutant Sensors: An optical sensor detects $PM_{2.5}$ using light scattering techniques to assess fine dust concentrations in the air.

3) Geolocation & Persistence: All data points are geo-tagged using a uBlox NEO-6M GPS module and stored locally on a microSD card to ensure integrity and avoid loss in low-connectivity areas.

4) Gaseous Pollutants: MP135 measures ozone (O_3) while MP503 detects formaldehyde (HCHO), both critical to air quality evaluation.

5) Platform & Dashboard: The platform, built with Next.js, TypeScript, and Tailwind CSS, features dynamic charts (Recharts) and maps (Leaflet). The backend handles data ingestion, APIs, and real-time AI predictions via containerized models.

TABLE II: Tools and Technologies Used in the Platform

Category	Technology/Tool	Purpose
Frontend	Next.js 14.2, TypeScript, Tailwind CSS	Modern, responsive, and type-safe web dashboard
UI Components	shadcn/ui, Radix UI, Lucide React	Accessible and interactive user interface
Data Visualization	Recharts, Leaflet	Charts, graphs, and geospatial maps
IoT Hardware	NodeMCU MEGA WiFi R3,$PM_{2.5}$ Sensor, UBlox NEO-6M GPS, MicroSD Module, MP135 & MP503	Mobile environmental sensing; data fields include Latitude, Longitude, CO, CO_2, Alcohol, Toluene, NH_4, Acetone, Dust, Voltage, and ADC readings
Data Analysis	Python (Pandas, NumPy, Scikit-learn)	Data cleaning, preprocessing, and feature engineering
Machine Learning	Scikit-learn, TensorFlow/Keras	Random Forest and Neural Network model development
Spatial Analysis	GeoPandas, DBSCAN	Geospatial data manipulation and clustering

C. Tools and Technologies

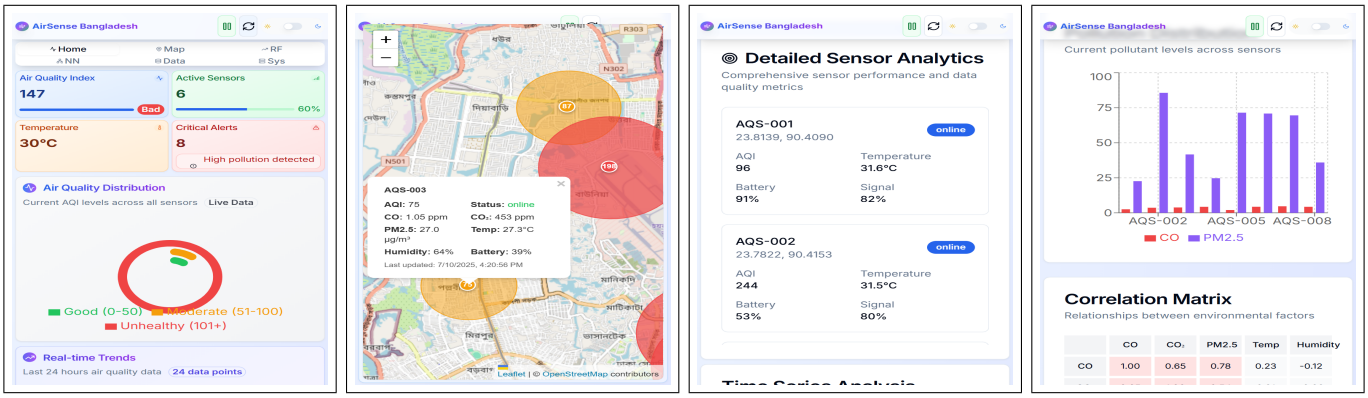
The key tools and technologies used in various components of the platform are summarized in Table II.

D. Data Collection and Preprocessing

1) Wireless Communication Protocols: The system utilizes Wi-Fi communication via the ESP8266 module embedded in the MEGA WiFi R3, enabling real-time data transmission from mobile IoT nodes to the cloud backend.

2) Data Acquisition Campaign: The IoT nodes were deployed on five public buses operating on major routes across Dhaka, Bangladesh, over three weeks. This campaign yielded a rich dataset of 10,234 geotagged records. The dataset captured a wide range of pollution levels. This rich dataset served as the foundational data for training and validating our models.

3) Preprocessing Pipeline: Raw sensor data was ingested by the backend and refined through a standardized pipeline.



(a) (b) (c) (d)

Fig. 3: UI snippets in smartphone view: (a) Dashboard and live metrics, (b) GPS-supported live data, (c) Detailed Sensor Analytics, (d) Live-pollutant levels.

During data cleaning, outliers were handled using the Interquartile Range (IQR) method. For feature engineering, timestamps were utilized to derive temporal features such as hour, day of the week, day of the year, and month, which helped capture daily and weekly pollution cycles.

E. Multi-Modal Analytical Engine Development

The core of the platform is a multimodal engine designed to extract spatial, predictive, and explanatory insights from mobile sensing data. Although established toolkits (Scikit-learn, TensorFlow) were employed for implementation, the principal machine-learning contributions are application-specific and concern architecture design, feature engineering, and validation tailored to mobile sensing. The main elements of the platform are customizable feature engineering that appropriately models the temporal dynamics of mobile sensors, an optimized multi-output feedforward neural network with three hidden layers of 128, 64, and 32 neurons for simultaneous multi-pollutant forecasting, and a detailed comparative performance evaluation claiming that the proposed model outperforms state-of-the-art models in this mobile sensing application. The multi-output neural network model was trained to be able to provide one-hour-ahead forecasting of pollutant concentrations at the geographic location of the sensor.

1) Spatial Analysis: Pollution Hotspot Identification: The DBSCAN algorithm was applied to geotagged data to detect pollution hotspots in Dhaka. Parameters *eps* and *min samples* were tuned iteratively. Identified hotspots are visualized on the dashboard's interactive map for actionable spatial insights.

2) Predictive Modeling: Forecasting Pollutant Levels: An RF classifier achieved 94.2% accuracy through ensemble decision trees with bootstrapping and feature randomness. For advanced predictions, the multi-output FNN achieved 96.8% accuracy using ReLU activation functions and a linear output layer for regression. This model drives next-hour air quality forecasts and predictive alerts.

3) Explanatory Analysis: Identifying Pollution Drivers: The RF model calculated Gini importance-based feature scores, quantifying each input variable's contribution to impurity reduction across all trees, revealing primary pollution drivers in an interpretable format.

4) Training and Evaluation Protocol: The dataset of 10,234 records was split into training (80%) and test (20%) sets using stratified sampling. We trained our proposed FNN and RF models on the training set and established baselines using Linear Regression and a Decision Tree trained under the same conditions. Model performance was then evaluated on the held-out test set, demonstrating the superior predictive accuracy of the RF and FNN models.

F. Evaluation Metrics

The performance of the analytical models was rigorously assessed using standard, well-established metrics to ensure their validity and reliability.

1) Predictive Model Validation: Model performance was measured using metrics appropriate to their specific tasks. Accuracy, the ratio of correctly predicted instances to the total instances, was used to evaluate the classification performance of the RF model. A combination of R-squared (R^2), Mean Absolute Error (MAE), which is the average of the absolute differences between prediction and actual observation, and Root Mean Squared Error (RMSE) (the square root of the average of squared differences between prediction and actual observation) was used for regression tasks to assess the forecasting accuracy of the FNN.

2) Spatial Cluster Validation: The output of the DBSCAN algorithm was validated qualitatively by mapping the identified clusters and confirming their overlap with known high-traffic corridors, industrial zones, and other areas of interest in Dhaka, confirming the model's real-world relevance.

IV. Results and Discussion

A. AI Model Performance Analysis

1) Predictive Accuracy: The predictive models were evaluated against classical baselines and an LSTM network for

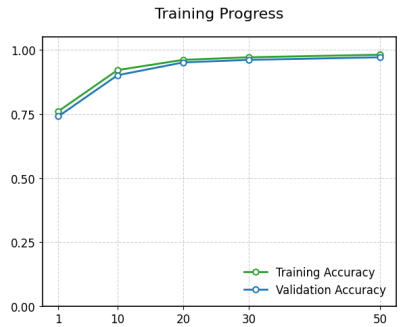


Fig. 4: FNN training and validation accuracy over epochs, demonstrating rapid convergence without significant overfitting

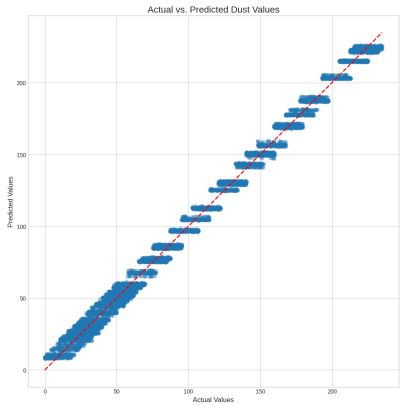


Fig. 5: Regression Plot for Dust Prediction

time-series forecasting. Table III demonstrates a clear performance hierarchy. The proposed FNN achieved the highest accuracy (R^2 = 0.96, Mean Absolute Error (MAE): 3.12, Root Mean Square Error (RMSE): 4.15), outperforming the LSTM network (R^2 = 0.95, MAE: 3.21, RMSE: 4.34) while requiring approximately four times less training time. The FNN's superior performance with significantly lower computational cost, as highlighted in Fig. 4, makes it optimal for resource-limited mobile IoT deployments requiring frequent retraining. The regression plots in Fig. 5 and Fig. 6 confirm the FNN's strong predictive power, demonstrating strong performance at a fraction of the computational cost.

2) Spatial Hotspot Discovery: Leveraging the unique mobility of our bus-mounted sensors, which provided extensive geographic coverage across Dhaka, we applied the DBSCAN

TABLE III: Comparative Model Performance on Test Set

Model	R^2	**MAE**	**RMSE**	**Train Time (s)**
Linear Regression	0.68	15.76	19.82	∼1
Decision Tree	0.85	8.91	11.23	∼3
Random Forest	0.94	4.52	5.68	∼25
FNN	**0.96**	**3.12**	**4.15**	**∼60**
LSTM	0.95	3.21	4.34	∼240

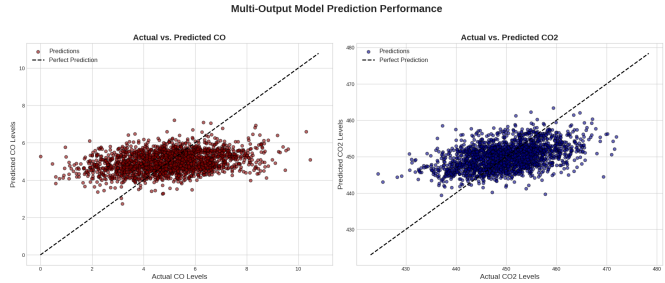


Fig. 6: Performance of the multi-output FNN for both CO and CO_2

algorithm to the collected geotagged data. This mobile approach enabled the identification of distinct pollution hotspots in Fig. 7, a city-wide spatial analysis not possible with a few fixed monitoring stations. For instance, identified hotspots correspond directly to major traffic intersections such as the Farmgate and Gulistan areas, as well as industrial zones in the Tejgaon region. This moves the analysis from simple observation to data-driven spatial intelligence, allowing for targeted policy interventions.

B. Discussion of Key Pollution Drivers

Table IV summarizes the relevance of the major air quality parameters to the sources of pollution and their impacts on human health [12].

C. System Implementation and Impact

The deployed system includes a real-time analytics dashboard with live sensor mapping, heatmaps, and time-series

TABLE IV: Air Quality Parameters and Health Impacts

Parameter	**Relevance**	**Health Impact on Humans**
CO (Carbon Monoxide)	vehicle exhausts and industrial activities	reduces oxygen-carrying capacity of blood; causes headaches, dizziness, unconsciousness, and at high levels, can be fatal.
Alcohol Vapors	solvents and chemical industries; emitted as VOCs	cause eye and respiratory irritation, dizziness, and long-term liver or neurological issues with prolonged exposure.
CO_2 (Carbon Dioxide)	product of combustion; elevated in poorly ventilated urban environments.	high indoor levels cause fatigue, dizziness, and reduced cognitive performance.
Toluene	common VOC from fuels, paints, and adhesives.	affects the nervous system, liver, and kidneys; causes fatigue, confusion, and skin/eye irritation.
NH_4 (Ammonium Ion)	Fertilizers, livestock waste, and industries.	irritates eyes, nose, throat, and lungs; dangerous at higher levels, causing respiratory burns.
Acetone	VOC from manufacturing and urban chemical usage.	irritates eyes, nose, and throat; may cause central nervous system depression.
DustVal (Dust Value)	$PM_{2.5}$/PM_{10} in the air.	causes asthma, lung inflammation, heart disease, etc.

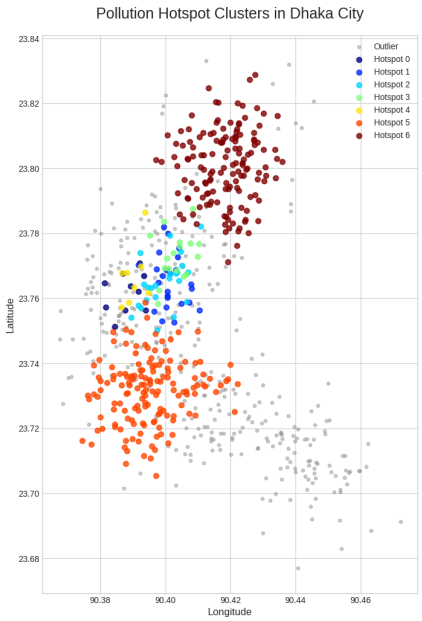


Fig. 7: Algorithmically identified pollution hotspot clusters, corresponding to major traffic intersections in Dhaka

pollutant trends, providing powerful analytics and monitoring capabilities for researchers and policymakers. The Real-world implementation addressed key engineering challenges: stability of the power supply through voltage regulator vehicle electrical systems, physical durability through robust weatherproof casing, and intermittent connectivity through SD card buffering for offline data storage, ensuring continuous operation under urban transport conditions.

D. Overall Discussion

The results show that low-cost mobile sensing combined with modern analytics enables effective urban air pollution monitoring and prediction. The high predictive performance supports a shift from reactive to proactive environmental management through alerts and dynamic policy actions. Unlike static monitors that offer limited spatial coverage, our mobile approach generates high-resolution pollution maps along major bus routes, revealing previously unseen pollution hotspots. This demonstrates the unique value of mobile sensing for data-driven urban policy and public health planning.

E. Limitations

To account for seasonal variations, the models must be retrained regularly. Besides, low-cost sensors may lack precision and require calibration using reference devices; future work will involve co-locating mobile nodes with certified, reference-grade monitors periodically to develop calibration models and improve data accuracy. The data is skewed toward major roadways while under-representing residential or low-traffic areas. To mitigate this roadway bias, future deployments could include placing devices on other municipal vehicles, such as waste management trucks following different routes, or deploying devices in key residential areas. Finally, the predictive models do not incorporate key meteorological data, such as wind speed and direction, which are dominant factors in pollution dispersion. Integrating these external variables is a critical area for future model enhancement.

V. Conclusion and Future Works

This paper presented AirSense, a validated, low-cost mobile sensing platform that leverages public buses in Dhaka to generate dynamic, high-resolution air quality forecasts. Our lightweight FNN, combined with DBSCAN clustering, provides a practical balance of accuracy and efficiency, delivering near-state-of-the-art performance ideal for scalable, real-time IoT deployment. The system offers significant advantages over static monitors, providing comprehensive spatial coverage at a fraction of the cost of traditional systems. Future work will focus on enhancing our models by incorporating meteorological data and exploring advanced spatial architectures. We also plan to scale the platform by integrating additional environmental sensors (e.g., for CO_2 and barometric pressure), adopting energy-efficient LoRa communication, and expanding the mobile sensor network to support smart city and public health initiatives further.

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